E1 213 Pattern Recognition and Neural Network
Assignment 1

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Phase 1: Bio-Optic Sensor Calibration

1.1 Gradient Descent vs. Closed Form

Ans: Ordinary least squares solution is implemented via the closed form solution $(X^TX)^{-1}Xy$ and also using the vectorized gradient descent algorithm with learning rate $\alpha = 0.01$ and number of iterations equal to 2000. The L2 norm difference between the two methods comes as 1.62×10^{-14}.

1.2 Primal to Dual Translation

Ans: The target dataset is threshold by median to give a binary dataset. The data is not linearly separable. So hard margin SVM will not be able to fit in the data. I used `scipy.optimize.minimize` to optimize, it is not able to converge and going on some infinite loop. Using `cvxopt.solvers.qp` was able to solve the problem but all the mu was very large suggesting that all of them are support vectors which is clearly not the case. All of these occurs as the optimization problem is not feasible.

Phase 2: Oncology Array

2.3 The Singularity Trap

Ans: Since $D > N$, so the rank of the matrix can be at best N and X^TX is a 1000×1000 matrix. So clearly, the matrix is singular. There is no `LinAlgError` in this case because of floating point approximation. But the high condition number of 9.66×10^{18} shows that the inverse is unstable and not reliable. Fig. 1a gives the condition number vs λ values plot, it can be seen with the increase of lambda there is a decrease in the condition number.

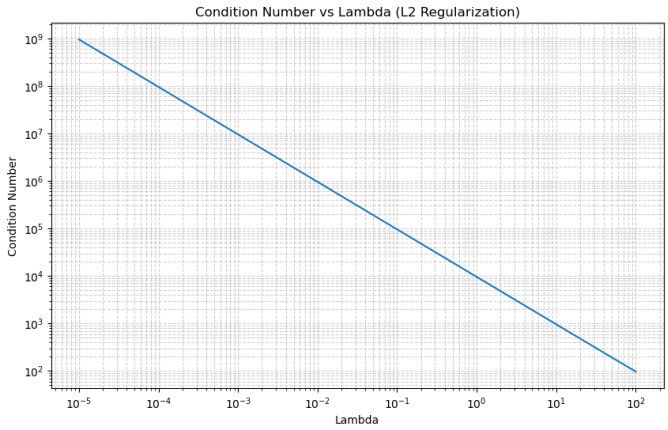
2.4 Lasso Sparsity

Ans: Fig. 1b shows the variation of the l_1 norm of weight value with $log(\lambda)$. The exact value of λ for which exactly 50% is equal to 0 is 0.37.

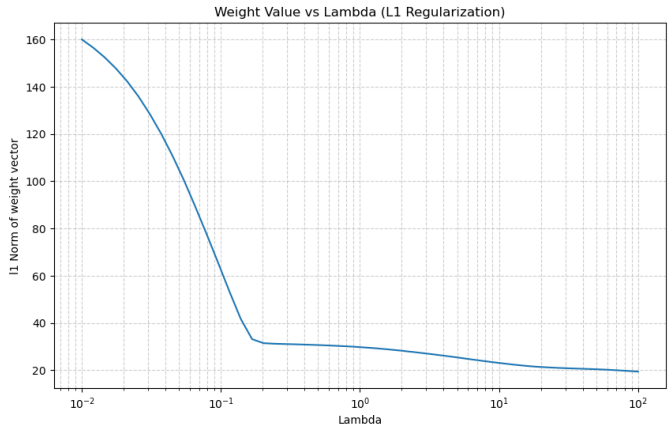
Phase 3: High Frequency ICU Telemetry

3.5 Vectorized Logistic Regression

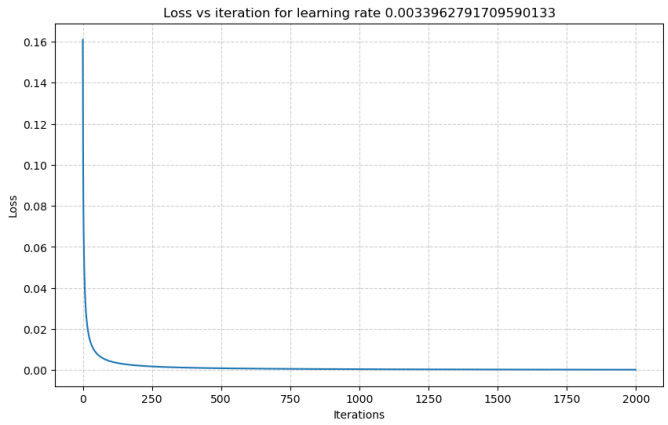
Ans: The Lipschitz smoothness factor for this case is very small $\frac{1}{L} = 0.003396$. So even when 2 or 3 times the factor is taken, the curve converges as the absolute value of the α is small. Only when very high value is taken then the curve does not converge. Fig. 2a gives the loss curve for a α that is close to the factor and Fig. 2b gives the loss curve for a very high α.



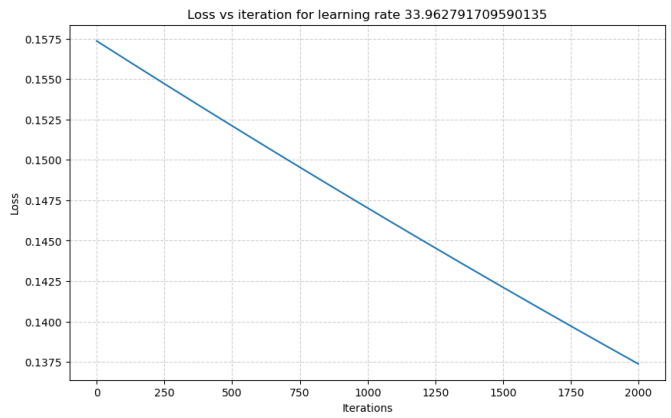
(a) κ vs λ for l_1 regularization



(b) l_1 norm of the weight vector vs $\log(\lambda)$



(a) Loss vs iteration curve for $\alpha = 0.00339$



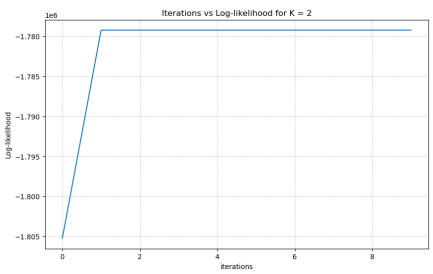
(b) Loss vs iteration curve for $\alpha = 33.96$

3.6 GMM Intialization & The E-Step

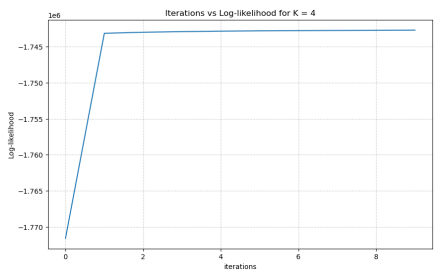
Ans: The initial log likelihood is -2857267.241 and after one iteration the log likelihood becomes -1771352.38.

3.7 EM Convergence and crashes

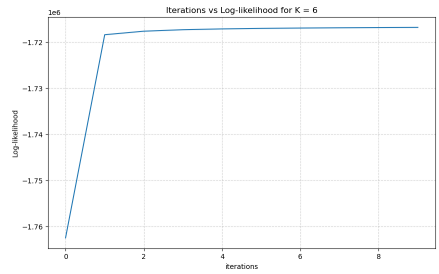
Ans: Fig. 3a, Fig. 3b and Fig. 3c gives the plots of Log-Likelihood vs Iteration for different values of K = 2,4 and 6. When the covariance matrix becomes singular, causing `numpy.linalg.LinAlgError` during inversion in the multivariate normal PD.



(a) Log-likelihood vs Iteration for $K = 2$



(b) Log-likelihood vs Iteration for $K = 4$



(c) Log-likelihood vs Iteration for $K = 6$

3.8 Decision Boundaries

Ans: Fig. 4a gives the plot of the decision boundary along with the Gaussians.

3.9 SVM with Slack KKT Verification

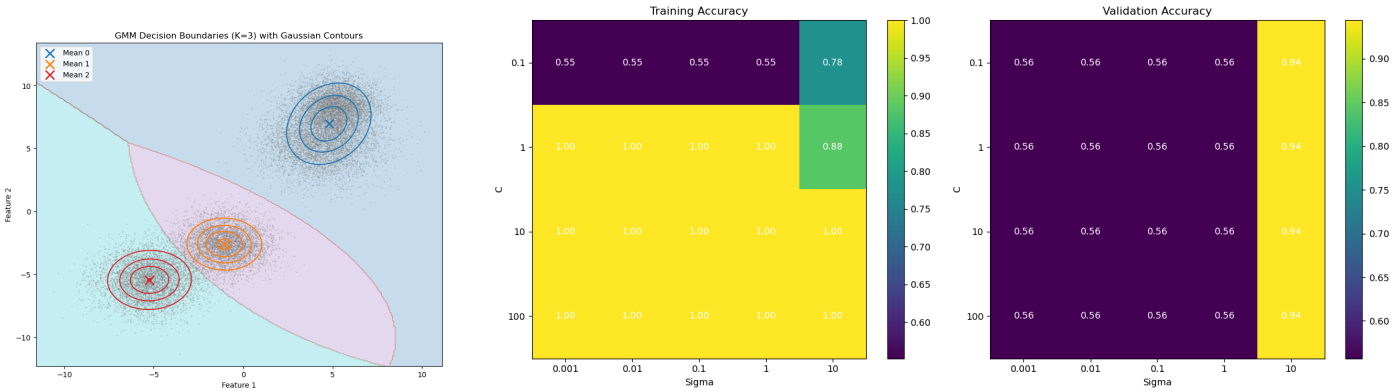
Ans: Index of point exactly on the margin 3374 and the value of mu for this is 0.0004879. Index of point classified safely outside the margin 8564 and the value of mu for this is 7.87×10^{-13}. No point has a μ close to 1. Highest μ is in the range pf 10^{-4} So, we can safely conclude that there is no point inside the margin.

3.10 Mercer's Condition

Ans: The Kernel matrix satisfy the Mercer's condition. The highest and the lowest eigenvalue is near 1 for small σ and when the σ becomes high, the matrix is ill conditioned but the lowest eigen value is always greater than 0. So, the matrix is PSD.

3.11 Hyperparameter Topography

Ans: Fig. 4b gives the heatmap of the search in the parameter space. Overfit occurs for all the $C \in \{1, 10, 100\}$ and $\sigma \in \{0.001, 0.01, 0.1, 1\}$. Only $\sigma = 10$ is able to fit the data properly.



(a) GM Decision Boundary

(b) Heatmap of the training and validation accuracies

Phase 4: Unscaled Electronic Health Records

4.12 Multiclass Extension & Scaling

Ans: The raw data does not converge at all the loss curve oscillates and the loop terminates at 2000 which is the max number of iterations preset. On the other hand, the scaled data converges in 1195 iterations where the tolerance is set to 10^{-6}.

4.13 Vectorization Benchmarking

Ans: The total time of running the Naive KNN is 25.78 secs, while the vectorized KNN takes only 0.53 secs. That is almost 50× faster.

4.14 Curse of Scale

Ans: The accuracy without the standard scalar is coming out to be 88.85% while the accuracy with the standard scalar is coming out to be 91.65%. Intuitively, that makes sense the unscaled data should give lower accuracy.

4.15 Bayes optimal & Underflow

Ans: The Naive Bayes classifier with log conditional is implemented on the whole dataset and it gives an accuracy of 91.31%. The underflow occurs at D=28 with `float32`.

Phase 5: Bias-Variance Decomposition

5.16 Empirical Bias-Variance via Bootstrapping

Ans: The bias of the $d = 1$ model and the $d = 15$ model are 2.33 and 2.31 respectively. While the variance is 0.001 and 0.01 respectively. This is expected the variance of the $d = 15$ model is higher while the bias is relatively less than $d = 1$ model. That is expected since the former model tries to fit in perfectly.

5.17 Frequentist vs. Bayesian Uncertainity

Ans: The frequentist variance is 0.0011 and the analytical bayesian variance is 0.0014. Fig. 5 gives the plot of both the distributions.
Frequentist: Expectation over repeated datasets $\mathcal{D}$:

$$\mathbb{E}_{\mathcal{D}}\left[\hat{\theta}\right], \qquad \mathrm{Var}_{\mathcal{D}}\left[\hat{\theta}\right] = \mathbb{E}_{\mathcal{D}}\left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}]\right)^2\right]$$

Bayesian: Expectation over the posterior $p(\theta \mid \mathcal{D})$:

$$\mathbb{E}_{\theta|\mathcal{D}}[\theta] = \int \theta\, p(\theta \mid \mathcal{D})\, d\theta, \qquad \Sigma_{\text{post}} = \left(\frac{X^\top X}{\sigma^2} + \frac{I}{\tau^2}\right)^{-1}$$

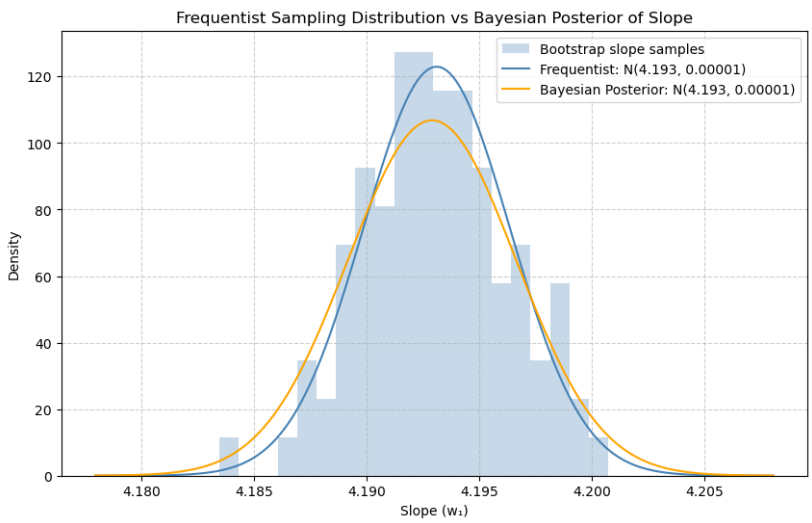


Figure 5: Distribution Function calculated analytically and with the frequentist approach